

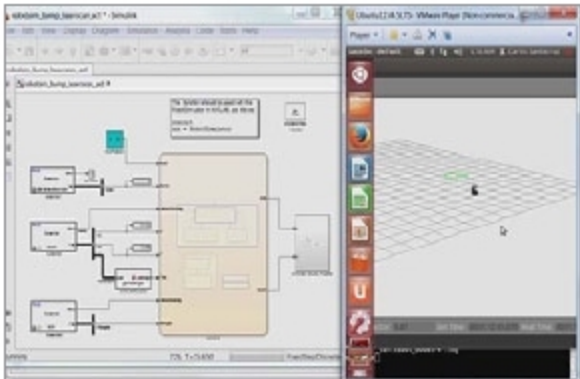
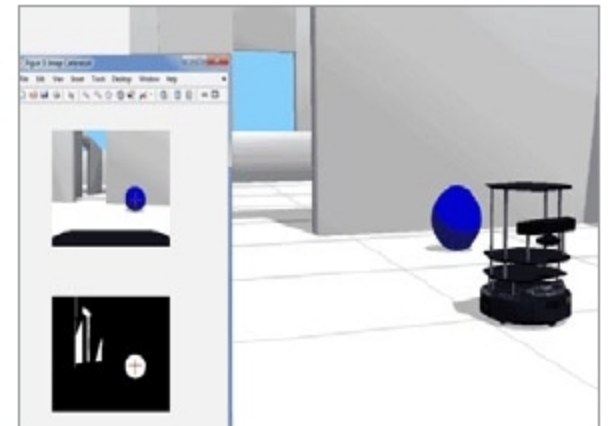
ROBOTICS And VR SOFTWARE For Enterprises And Developers

Robot Operating System

Visit: www.ros.org

Full version: Free (BSD licence)

Robot Operating System (ROS) is a framework for developing robotics applications. It is a collection of tools, libraries and conventions, which provides the platform to create robotics programs. Core components of ROS are communication middleware, some robot-specific features and command-line tools. The communication interface takes care of message passing, recording, playback, procedure calling and distributed parameter systems. Robot-specific services include Robot Description Language support, Robot Geometry Library, Standard Robot Messages, Pose Estimation and Localisation, diagnostics and troubleshooting. Commands and instructions are provided through rviz (command-line tool) and rqt (qt-based graphical interface). Latest version of the software is called Melodic Morenia. ROS runs on Linux (Ubuntu Artful, Bionic and Devian Stretch). It is also experimentally compatible with MacOS.



Robotics System Toolbox

Visit: <https://in.mathworks.com>

Full version: Paid

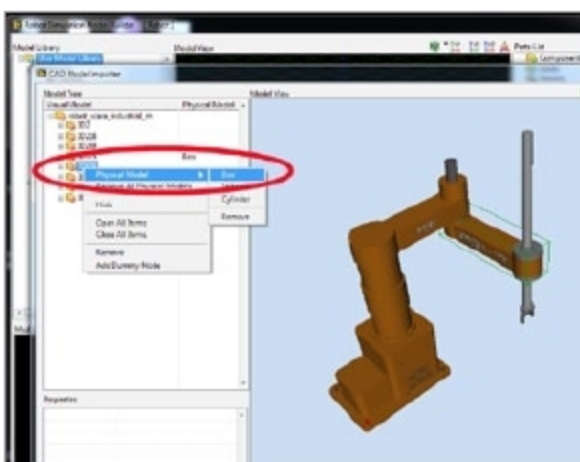
MathWorks has developed Robotics System Toolbox, which creates an interface between ROS and MATLAB, or ROS and Simulink platforms. The complete environment can be used to test and evaluate robotic functions on actual ROS-driven robots as well as on robot simulation tools (like Gazebo). The Toolbox can be used to create various algorithms like path planning and following of robot, obstacle avoidance, scanning and scan matching, state estimation, bi-directional communication as well as more advanced logics like inverse kinematics, kinematic constraints and rigid body tree-based dynamics programming. It is a great tool for various autonomous robotic applications like humanoid robots, aerial vehicles, ground robots and manipulators.

RoboLogix

Visit: www.robologix.com

Full version: Paid

RoboLogix is a versatile industrial robot simulator software, beneficial for students, new engineers as well as professionals. The platform represents a digital five-axis industrial robot that allows you to run, test, debug and teach robotics programs. Its interface is highly interactive and interesting with 3D visual graphics. The learning curve is not too complicated because of the easy-to-understand instruction sets. With RoboLogix, you can assess and evaluate your industrial robotic applications digitally with accuracy before implementing these in real life. Various applications like pick-and-place, palletising, welding and painting can be programmed with RoboLogix.



LabVIEW

Visit: www.ni.com/robotics

Full version: Paid

National Instruments (NI) developed Laboratory Visual Instrument Engineering Workbench (LabVIEW) as a system design platform and Visual Programming Language development environment. It can be actively used for robotics applications. Algorithms for various functions like sensor communication, motion path, obstacle avoidance, steering and kinematics, among others, can be achieved with LabVIEW. The software is compatible across multiple platforms, namely, Windows, MacOS, Linux and Unix.

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Browser: Google Chrome
(www.google.com/chrome)

PDF Reader: Foxit
(www.foxitsoftware.com)

Media player: VLC (www.videolan.org)